MILL TEST CERTIFICATE

AXIS
PIPE AND TUBE

XIS PIPE AND TUBE, LLC.
451 Louis E. Mikulin Rd.
ryan TX 77807

Customer:		**Sales Order:**	**Invoice No.:**
MRC Global		0630004951	1600049629
PO Box		**PO:**	**Date of Issue:**
Charleston,WV,25322		S 7AA 637379	07/23/2024
877-294-7574			
Certificate No.:	1600049629-24		
Product Description:	5L 12.750 0.500 65.48# X52BDRH5BW-20P2		
Specification:	5L 46th Edition May 2019		

Heat Number	Lot Number	Mat Number	Steel Making / Coil Rolling
B402912	LT24185028	332758	Steel Dynamics Columbus, LLC , Columbus - MS - US / Steel Dynamics Columbus, LLC , Columbus - MS - US

Mechanical Properties

Tensile Test - Strip Specimen Gauge Length 2"								Temp (F) -20 CVN								Temp (F) -20										
Loc. / Orien	Size	0.5%EUL Y.S(psi)	U.T.S. Body (psi)	U.T.S. Weld (psi)	Y/T Ratio	E.L. (%)	Hardness (HRB)	Body (ft-lbf)				Shear Area (%)				Weld (ft-lbf)				Shear Area (%)				Size	Loc. / Orien	Acceptance Criteria (ft-lbf)(AVG)
								1	2	3	Avg	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg			
T180°	1 1/2"	66500	74000	73100	0.90	41.8	83	205	201	199	202	100	100	100	100	98	153	179	144	95	95	100	97	10x7.5 mm	3T90	15
								Lateral Expasion (in)				UB	UB	UB	UB	Lateral Expasion (in)				0.101	0.088	UB	0.094			

Analysis (H/P)	Chemical Composition (Max %)																		
	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEpcm (API Max. A.C.)	CE Results	
H	0.06	0.220	0.82	0.013	0.003	0.05	0.07	0.010	0.14	0.002	0.019	0.026	0.015	0.0003	0.0016	0.0070	0.25	0.12	
P	0.06	0.192	0.77	0.010	0.001	0.06	0.06	0.010	0.12	0.002	0.013	0.022	0.013	0.0002	0.0021	0.0078	0.25	0.12	
P	0.06	0.193	0.78	0.010	0.001	0.06	0.06	0.010	0.12	0.002	0.013	0.022	0.013	0.0002	0.0021	0.0082	0.25	0.12	

Following Controls / Tests have been satisfactorily performed:
1.- Hydrostatic Testing: 4,090.00 psi Pressure for 5.00 seconds
2: Ultrasonic Tested to N10 - passed
3: Seam Annealed min 1,600 F
4: Flattening Test Passed

Remarks:
1: This material meets:Grades ASTM A53/A53M-22 Grade B
API 5L HFW PSL2 X52M,
2: This product meets the requirements of ASME BPVC.II.A Specification SA-53B-2023
3: Made and Melted in USA
4: Date of Manufacture 07/2024
5: This product is silicon and/or aluminum fully killed.
6: Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1.B)
7: This material meets the requirements of NACE MR0175-2021 / ISO 15156:2020

Acronyms:
L = Longitudinal
Y.S. = Yield Strength
E.L. = Elongation
P = Product Analysis
UB = Un-Broken
T = Transverse
U.T.S. = Ultimate Tensile Strength
H = Heat Analysis
A.C. = Acceptance Criteria
N/A = Not Applicable

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

Quality Manager